

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN STACK

Practical 2

Roll No.: T013	Name: Saqlain Khan
Class: TYIT	Batch: 1
Date of Assignment: 30-06-2026	Date/Time of Submission: 30-06-2026

AIM :- Core Javascripts Concepts for MERN

A) 1. Create functions and call them with parameters.

CODE :-

```
function calculateSalary(name,basicSalary) {  
    let bonus = 5000;  
    let totalSalary = basicSalary + bonus;  
    console.log("Employee Name : ",name);  
    console.log("Basic Salary : ",basicSalary);  
    console.log("Bonus : ",bonus);  
    console.log("Total Salary : ",totalSalary);  
}  
  
calculateSalary("Saqlain Khan",35000);
```

OUTPUT :-

```
● PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2A.js  
Employee Name : Saqlain Khan  
Basic Salary : 35000  
Bonus : 5000  
Total Salary : 40000
```

2. CODE :-

```
function generateBill(product, quantity, price)  
{  
    let total = quantity * price;
```

```
let bill = total + (total * 0.09);

console.log("Product : ", product);

console.log("Quantity : ", quantity);

console.log("Price Per Item : ", price);

console.log("Total Bill : ", bill);

}

generateBill("Laptop",2,50000);
```

OUTPUT :-

```
PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2A.js
Product : Laptop
Quantity : 2
Price Per Item : 50000
Total Bill : 109000
```

B) Works with Arrays and Object in Javascripts

CODE :-

```
let fruits = ["Apple", "Banana", "Guava"];

let Student = {

    Name : "Saqlain Khan",

    Age : 20,

    City : "Mumbai"

};

console.log("Fruits : ", fruits);

console.log("Second Fruit : ", fruits[1]);

console.log("Student Name : ", Student.Name);

console.log("Student Age : ", Student.Age);

console.log("Student City : ", Student.City);
```

OUTPUT :-

```
● PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2B.js
Fruits : [ 'Apple', 'Banana', 'Guava' ]
Second Fruit : Banana
Student Name : Saqlain Khan
Student Age : 20
Student City : Mumbai
```

C) Demonstrate use of arrow functions.

CODE :-

```
let add = (a,b) => {
    return a + b;
};

let sub = (a,b) => {
    return a - b;
};

let mul = (a,b) => {
    return a * b;
};

let div = (a,b) => {
    return a / b;
};

console.log("Addition = ", add(10, 20));
console.log("Subtraction = ", sub(10, 20));
console.log("Multiplication = ", mul(10, 20));
console.log("Division = ", div(10, 20));
```

OUTPUT :-

```
● PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2C.js
Addition = 30
Subtraction = -10
Multiplication = 200
Division = 0.5
```

D) Write a program using array methods (map, filter, reduce).

CODE :-

```

let numbers = [10,20,30,40,50];

let doubled = numbers.map(num => num * 2);

let greater25 = numbers.filter(num => num > 25);

let total = numbers.reduce((sum,num) => sum + num, 0);


console.log("Original Array : ",numbers);

console.log("Map : ",doubled);

console.log("Filter : ",greater25);

console.log("Reduce : ",total);

```

OUTPUT :-

```

PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2D.js
Original Array : [ 10, 20, 30, 40, 50 ]
Map : [ 20, 40, 60, 80, 100 ]
Filter : [ 30, 40, 50 ]
Reduce : 150

```

E) Create a program that manipulates objects and displays output.

CODE :-

```

let student = {

    rollNo: 13,

    Name: "Saqlain Khan",

    Marks: 67

};


console.log("\n Accessing Properties: ");

console.log("Name : ", student.Name);

console.log("Marks : ", student.Marks);


// Updating Property

student.Marks = 69;

```

```

console.log("\n After Updating Marks : ");

console.log(student);

// Adding a Property
student.city = "Mumbai";

console.log("\nAfter Adding City : ");

console.log(student);

// Deleting
delete student.rollNo;

console.log("\nAfter Deleting : ");

console.log(student);

```

OUTPUT :-

```

PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2E.js

  Accessing Properties:
  Name :  Saqlain Khan
  Marks :  67

  After Updating Marks :
  { rollNo: 13, Name: 'Saqlain Khan', Marks: 69 }

  After Adding City :
  { rollNo: 13, Name: 'Saqlain Khan', Marks: 69, city: 'Mumbai' }

  After Deleting :
  { Name: 'Saqlain Khan', Marks: 69, city: 'Mumbai' }
PS C:\Users\mvluc\Desktop\MERN Practical> 

```

- F) Implement a small JavaScript program combining functions, arrays, and objects.

CODE :-

```

let students = [

  {

    rollNo: 13,

```

```

        Name: "Saqlain Khan",
        Marks: 67
    },
    {
        rollNo: 14,
        Name: "Ritesh",
        Marks: 68
    },
    {
        rollNo: 15,
        Name: "Himani",
        Marks: 69
    }
];

function display(s)
{
    console.log("Student Details");
    console.log("-----");
    for (let stud of s)
    {
        console.log(`Roll No : ${stud.rollNo}`);
        console.log(`Name : ${stud.Name}`);
        console.log(`Marks : ${stud.Marks}`);
        console.log(`-----`);
    }
}

display(students);

```

OUTPUT :-

```
-----  
PS C:\Users\mvluc\Desktop\MERN Practical> node Prac2F.js  
Student Details  
-----
```

```
Roll No : 13  
Name : Saqlain Khan  
Marks : 67  
-----
```

```
Roll No : 14  
Name : Ritesh  
Marks : 68  
-----
```

```
Roll No : 15  
Name : Himani  
Marks : 69
```